

Vendor Inventory and Spec Delta — AutoCAD 2026 / BricsCAD V26

Phase 1 of the autolisp-spec exhaustiveness pass

Codex

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1 1 Purpose

This document is the Phase-1 deliverable of the ongoing work to bring AutoLISP Spec to exhaustive coverage of the AutoLISP / Visual LISP language as shipped by Autodesk AutoCAD 2026 and Bricsys BricsCAD V26.

It records:

- the canonical vendor symbol inventories that were extracted from AutoCAD 2026 and BricsCAD V26 vendor documentation,
- the per-symbol availability tags (AutoCAD-only / BricsCAD-only / both) at the level the vendor docs make explicit,
- the delta between those vendor inventories and the existing specification draft,
- a categorised list of missing entries that subsequent phases will fill in.

This document is **not** a normative part of the specification.

It is a working artefact tracking what the spec covers and what it still needs to cover.

2 2 Scope and Method

2.1 2.1 Scope

In scope for the inventory:

- core AutoLISP and Visual LISP functions, special forms, and reader-syntax classes,
- DCL (Dialog Control Language) functions used from AutoLISP,
- ActiveX / COM bindings (`vla*-*`), reactors (`vlr*-*`), and the Visual LISP family (`vl*-*`),
- Express Tools functions (`acet*-*`) and `acad*-*` / `acad_*` helpers documented in the AutoLISP Reference,
- BricsCAD-specific extensions enumerated on the BricsCAD-Specific LISP Functions page,
- per-version flags where the vendor explicitly provides them.

Out of scope for this inventory (per project decision dated 2026-04-26):

- BricsCAD BIM, Mechanical, GeoCoordinateSystem, and SheetSet-Manager LISP APIs.
- Vertical / discipline-specific extensions are listed only as reference URLs.

2.2 2.2 Method

1. Fetched the AutoCAD 2026 "AutoLISP Functions By Name and Feature" index page.
2. Fetched all 24 alphabetical sub-indexes (Non-alphabetic, A-Z except J/K/Y) and enumerated every named entry.
3. Fetched the AutoCAD 2026 feature-category index links to confirm coverage of Express Tools, ActiveX, DCL, Reactors, and Selection/Symbol-Table groups.
4. Fetched the BricsCAD V26 BricsCAD-Specific LISP Functions page, the V26 Generic Properties Functions page, the V26 VLX Namespace Support page, the V26 read+write text files page, the V26 `vlisp-compile` page, the V26 Optimised Code Patterns page, and the V26 LISP Optimiser pages.
5. Cross-checked against the V25 "What's New in BricsCAD Lisp" page and the V14/V9 LISP Compatibility pages where current pages returned 404.
6. Normalised every symbol name to lowercase for diffing.
7. Computed the diff against the 328 dictionary entries already present in the specification draft.

2.3 2.3 Caveats

- The Bricsys developer-reference index page is JavaScript-driven, so raw HTTP fetching of the framed TOC returns only the top-level category names. The full TOC tree is built at runtime from a `chmtop.c2wtopf.TITEMS` JavaScript array. Phase 1 was therefore re-run inside a real Chromium instance (via the Claude-in-Chrome MCP) on 2026-04-26: the live TITEMS tree was walked recursively and dumped to `bricscad-lisp-toc.txt` (1934 entries, 95.6 KB). This is the authoritative BricsCAD V26 LISP catalogue.
- The historical compatibility-table pages (`IDR_LISP_B100.htm`, `CompatibilityWithAutoCAD.htm`, `LISPCompatibility.htm`) returned 404 in V25 and V26; only the V9 page is currently reachable. Per-symbol BricsCAD support for AutoCAD-documented AutoLISP names is now established by direct presence in the BricsCAD V26 TOC, not inferred — see §6 and §9.

- AutoCAD’s localised reference URLs (FRA, KOR, CHT, ENU, ...) all resolve to the same content keyed by GUID. This inventory records ENU URLs only.

3 3 Authoritative Source URLs

3.1 3.1 AutoCAD 2026

- AutoLISP Functions By Name and Feature index: <https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD-AutoLISP-Reference/files/GUID-4CEE5072-8817-4920-8A2D-7060F5E16547.htm>
- Per-letter alphabetical indexes (Non-alphabetic, A, B, C, D, E, F, G, H, I, L, M, N, O, P, Q, R, S, T, U, V, W, X, Z) — see GUID table in §6.2 of this document.

3.2 3.2 BricsCAD V26

The BricsCAD V26 LISP TOC (recovered live from the JS-built tree on 2026-04-26) has 30 top-level sections under the LISP root. Section headings, in TOC order, with their entry pages:

Section	Page
LISP overview	source/LISP.htm
Introduction	source/Introduction.htm
Compatibility Changes	source/CompatibilityChanges.htm
Application Security	source/ApplicationSecurity.htm
AutoLISP Standard Functions	source/AutoLISPStandardFunctions.htm
Automatic LISP + VBA Loading	source/AutomaticLISPVBALoading.htm
Generic Properties Functions	source/GenericPropertiesFunctions.htm
BricsCAD DCL Support	source/BricsCADDCLSupport.htm
BricsCAD LISP Memory Configuration	source/BricsCADLISPMemoryConfiguration.htm
BricsCAD LISP Error Control	source/BricsCADLISPErrorControl.htm
BricsCAD LISP Functions	source/BricsCADLISPFunctions.htm
BricsCAD LISP Protected Names	source/BricsCADLISPProtectedNames.htm
BricsCAD BIM LISP API	source/BricsCADBIMLISPAPI.htm
BricsCAD Civil3D LISP API	source/BricsCADCivil3DLISPAPI.htm
BricsCAD Civil GIS LISP API	source/BricsCADCivilGISLISPAPI1.htm
BricsCAD Civil Tin Surface LISP API	source/BricsCADCivilTinSurfaceLISPAPI.htm
BricsCAD GeoCoordinateSystem LISP API	source/BricsCADGeoCoordinateSystemLISPA.htm
BricsCAD Mechanical LISP API	source/BricsCADMechanicalLISPAPI.htm
BricsCAD SheetSetManager COM / LISP API	source/BricsCADSheetSetManagerCOMLISPAP.htm
BricsCAD SheetMetal LISP API	source/BricsCADSheetMetalLISPAPI.htm
Extended AutoLISP Functions	source/ExtendedAutoLISPFunctions.htm
VLE LISP Function Library	source/VLELISPFunctionLibrary.htm
VLX NameSpace Support	source/VLXNameSpaceSupport.htm
ExpressTools API Support	source/ExpressToolsAPISupport.htm
DOSLib for Linux + Mac	source/DOSLibforLinuxMac.htm
LISP "Fast-COM"	source/LISPFastCOM.htm
LISP Optimiser	source/LISPOptimiser.htm
LISP Benchmark	source/LISPBenchmark.htm
General Performance Hints	source/GeneralPerformanceHints.htm
DEScoder LISP encryption	source/DEScoderLISPencryption.htm
Support for UltraEdit Editor	source/SupportforUltraEditEditor.htm

The vertical-API sections (BIM, Civil3D, Civil GIS, Civil Tin Surface, GeoCoordinateSystem, Mechanical, SheetSetManager, SheetMetal) are out of scope for `clautolisp` per project decision dated 2026-04-26 and are listed only as reference URLs.

Construct individual page URLs as https://developer.bricsys.com/bricscad/help/en_US/V26

4 4 Inventory Summary

The 2026-04-26 Phase-1-revisited counts (after walking the BricsCAD V26 TOC inside Chrome) are:

Set	Count
AutoCAD 2026 documented symbols	485
BricsCAD V26 in-scope LISP entries (TOC leaves with .htm pages, verticals removed)	941
- Standard AutoLISP Functions section	460
- BricsCAD-Specific LISP Functions section	86
- Generic Properties Functions section	6
- BricsCAD LISP Error Control section	4
- Extended AutoLISP Functions section	3
- VLE LISP Function Library section	117
- VLX NameSpace Support section	13
- ExpressTools API Support section	261
- DOSLib for Linux + Mac section	23
Existing specification dictionary entries (post Phase 3)	594
Spec BricsCAD V26 (already present)	564
In BricsCAD V26 TOC but missing from the spec (Phase-2 follow-up)	377
In spec only (not on either vendor index — concept entries)	14

The "In spec only" set is mostly concept-level entries that have no single vendor function page (reader-syntax classes, type entries, the constants `nil`, `t`, `pi`, `pause`). One entry, `autoload`, was a typo of `autoload` and was renamed in Phase 2.

The "377 missing" set is the Phase-2 follow-up: it expands chapter 24 (BricsCAD Extensions) to cover the BricsCAD-only sections that the JS-driven TOC was hiding from the original Phase-1 fetch. The breakdown is in §8.13.

5 5 Per-Symbol Availability Model

For Phase 2 onward, every dictionary entry in the spec must carry an explicit availability cell. The agreed-upon vocabulary is:

- **AutoCAD**: documented on Autodesk's per-symbol page; not documented by Bricsys.
- **BricsCAD**: documented by Bricsys; not documented by Autodesk.

- **Both:** documented (or strongly implied as compatible) by both vendors.
- **Both (presumed):** in the AutoCAD AutoLISP reference and not contradicted by Bricsys, with no direct Bricsys page; default for AutoLISP core symbols pending a per-symbol confirmation.
- **AutoCAD-Windows:** documented as Windows-only by Autodesk (ActiveX, `acet-*` Express Tools, several DCL helpers); BricsCAD status TBD.
- **Both with divergence:** documented by both vendors, but with different signature, return shape, or platform footprint.

In the dictionary entries this becomes:

```
#+beginexample
```

5.0.1 Availability

- AutoCAD 2026: yes (Windows + macOS, except where noted).
- BricsCAD V26: yes (Windows + macOS + Linux).
- Status: documented in both.

```
#+endexample
```

6 6 AutoCAD 2026 Inventory

6.1 6.1 By family

- Arithmetic, comparison, bitwise: `+`, `-`, `*`, `/`, `=`, `/=`, `<`, `<=`, `>`, `>=`, `~`, `1+`, `1-`, `abs`, `atan`, `boole`, `cos`, `exp`, `expt`, `fix`, `float`, `gcd`, `log`, `logand`, `logior`, `lsh`, `max`, `min`, `minusp`, `numberp`, `rem`, `sin`, `sqrt`, `zerop`.
- Equality, predicates, type: `eq`, `equal`, `atom`, `boundp`, `listp`, `null`, `not`, `type`.
- Conditionals and binding: `if`, `cond`, `and`, `or`, `progn`, `repeat`, `while`, `foreach`, `setq`, `set`, `lambda`, `defun`, `defun-q`, `defun-q-list-ref`, `defun-q-list-set`, `function`, `quote`, `apply`, `eval`.
- List manipulation: `car`, `cdr`, `cadr`, `caddr`, `cons`, `list`, `append`, `length`, `last`, `mapcar`, `member`, `nth`, `reverse`, `subst`, `assoc`.

- Symbol handling: `atoms-family`, `snvalid`, `vl-symbol-name`, `vl-symbol-value`, `vl-symbolp`.
- String handling: `ascii`, `chr`, `strcase`, `strcat`, `strlen`, `substr`, `wcmatch`, plus `vl-string-*` family (`vl-string-elt`, `vl-string-left-trim`, `vl-string-mismatch`, `vl-string-position`, `vl-string-right-trim`, `vl-string-search`, `vl-string-subst`, `vl-string-translate`, `vl-string-trim`, `vl-string->list`, `vl-list->string`).
- Numeric / string conversion: `atof`, `atoi`, `itoa`, `rtos`, `angtof`, `angtos`, `distof`, `cvunit`.
- Geometric: `angle`, `distance`, `inters`, `osnap`, `polar`, `textbox`, `trans`.
- Selection sets: `ssadd`, `ssdel`, `ssget`, `ssgetfirst`, `sslenght`, `ssmemb`, `ssname`, `ssnamex`, `sssetfirst`.
- Entity / drawing: `entdel`, `entget`, `entlast`, `entmake`, `entmakex`, `entmod`, `entnext`, `entsel`, `entupd`, `handent`, `nentsel`, `nentselp`, `namedobjdict`, `regapp`, `tblnext`, `tblobjname`, `tblsearch`, `xdroom`, `xdsize`, `dictadd`, `dictnext`, `dictremove`, `dictrename`, `dictsearch`, `getpropertyvalue`, `setpropertyvalue`, `ispropertyreadonly`, `dumpallproperties`.
- File and path: `close`, `findfile`, `findtrustedfile`, `load`, `open`, `read-char`, `read-line`, `write-char`, `write-line`, `vl-directory-files`, `vl-file-copy`, `vl-file-delete`, `vl-file-directory-p`, `vl-file-rename`, `vl-file-size`, `vl-file-systime`, `vl-filename-base`, `vl-filename-directory`, `vl-filename-extension`, `vl-filename-mktemp`, `vl-mkdir`, `vl-get-resource`.
- Display / printer: `prin1`, `princ`, `print`, `prompt`, `terpri`, `textpage`, `textscr`, `graphscr`, `grclear`, `grdraw`, `grread`, `grtext`, `grvecs`, `redraw`, `showhtmlmodalwindow`, `tablet`, `vector_image`, `vports`, `setview`.
- User input / Command: `command`, `command-s`, `initcommandversion`, `initdia`, `initget`, `getangle`, `getcorner`, `getdist`, `getfiled`, `getint`, `getkword`, `getorient`, `getpoint`, `getreal`, `getstring`, `menucmd`, `menugroup`, `startapp`, `vl-cmdf`.
- DCL: `load_dialog`, `unload_dialog`, `new_dialog`, `start_dialog`, `done_dialog`, `term_dialog`, `action_tile`, `client_data_tile`, `mode_tile`, `set_tile`, `get_tile`, `get_attr`, `dimx_tile`, `dimy_tile`, `start_image`, `end_image`, `fill_image`, `slide_image`, `vector_image`, `start_list`, `end_list`, `add_list`.

- Application / system: `alert`, `exit`, `quit`, `help`, `setfunhelp`, `ver`, `setvar`, `getvar`, `getenv`, `setenv`, `getcfg`, `setcfg`, `getcname`, `layoutlist`, `acdmenableupdate`, `arx`, `arxload`, `arxunload`, `autoarxload`, `autoload`, `vl-load-all`, `vl-load-com`, `vl-load-reactors`, `vl-doc-export`, `vl-doc-import`, `vl-doc-ref`, `vl-doc-set`, `vl-bb-ref`, `vl-bb-set`, `vl-propagate`, `vl-acad-defun`, `vl-acad-undefun`, `vl-arx-import`, `vl-list-exported-functions`, `vl-list-loaded-vlx`, `vl-vlx-loaded-p`, `vl-unload-vlx`, `vl-vbaload`, `vl-vbarun`, `vlisp-compile`, `vl-list-length`, `vl-list*`, `vl-prin1-to-string`, `vl-princ-to-string`, `vl-registry-delete`, `vl-registry-descendents`, `vl-registry-read`, `vl-registry-write`.
- Memory and debugging: `alloc`, `expand`, `gc`, `mem`, `trace`, `untrace`.
- Error handling: `*error*`, `*pop-error-mode*`, `*push-error-using-command*`, `*push-error-using-stack*`, `vl-catch-all-apply`, `vl-catch-all-error-message`, `vl-catch-all-error-p`, `vl-exit-with-error`, `vl-exit-with-value`.
- Layer states: `layerstate-addlayers`, `layerstate-compare`, `layerstate-delete`, `layerstate-export`, `layerstate-getlastrestored`, `layerstate-getlayers`, `layerstate-getnames`, `layerstate-has`, `layerstate-import`, `layerstate-importfromdb`, `layerstate-removelayers`, `layerstate-rename`, `layerstate-restore`, `layerstate-save`.
- Higher-order list: `vl-every`, `vl-some`, `vl-position`, `vl-remove`, `vl-remove-if`, `vl-remove-if-not`, `vl-member-if`, `vl-member-if-not`, `vl-cons`, `vl-sort`, `vl-sort-i`.
- ActiveX / COM (Windows): `vlax-*` family — see §6.2 for the full list extracted from the V Functions page.
- Reactors: `vlr-*` family — see §6.2 for the full list.
- Express Tools (Windows-only): `acet-*` family of 50 entries, plus `acad_colordlg`, `acad_helpdlg`, `acad_strlsort`, `acad_truecolorcli`, `acad_truecolordlg`, `acad-pop-dbmod`, `acad-push-dbmod`.

6.2 6.2 Feature categories (canonical Autodesk grouping)

This is the Autodesk-published partition. Phase 2 should align spec chapter ordering with these categories where possible, since this is how vendor users locate functions.

Group	GUID (page)
Application-Handling	GUID-E2B683FE-43FB-4F5D-A9AE-809772FE8D31
Arithmetic	GUID-FC14467C-63B9-4400-8C6A-266D21C846AE
Equality and Conditionals	GUID-C6A7FC58-7A07-4650-9D59-12CAFD194FDA
Error-Handling	GUID-D114F1C4-DE8E-422D-8F3F-C43707B8212B
Function-Handling	GUID-6804DF2D-E5C4-466A-9869-3CDD7A031FAA
List Manipulation	GUID-00A75AFB-5708-4D0C-B18C-F74DDC4D65F5
String-Handling	GUID-8543549B-F0D7-43CD-87A3-F6B827FF0B88
Symbol-Handling	GUID-796EFF94-E904-4F5B-B120-C6D2DCD8646A
Data Conversion	GUID-5BDBD4BC-289E-4A49-9BB9-DF6A6147E0F5
Device Access	GUID-018821E0-23F6-43F0-9548-6ABC70A180A0
Display Control	GUID-9C2C56A3-0BEF-45AB-A7FE-5825339DCB41
File-Handling	GUID-F70DECFC-DBE1-4F04-A64C-B3F869A636A1
Geometric	GUID-8C29D7A3-9CBB-47A8-A936-87118782303B
Query and Command	GUID-F9ABE09D-9A71-4C00-8E5D-5BA70C3797B1
User Input	GUID-49DBE7DD-D1D2-4B3A-9718-E3F4E2FD5E81
Memory Management	GUID-0EA35DD5-0C92-4AF8-B3AE-EFA35A5FDF51
Windows Registry / Property List Files	GUID-E3D3DA29-AF63-41D3-865C-39E50FF31596
Extended Data-Handling	GUID-0949AEA9-3942-42C5-8DF4-BC58758319BD
Object-Handling	GUID-1197B695-C73C-4B4D-AD5D-8F8EB64FB251
Selection Set Manipulation	GUID-43715289-5715-4F96-8D48-3169FA26777A
Symbol Table and Dictionary-Handling	GUID-7AACB386-1AA1-4BDF-B5D7-A1E23A51CA11
DCL Dialog Box Opening and Closing	GUID-A96C28B5-4B9A-4EE8-8C46-808F14217777
DCL Tile- and Attribute-Handling	GUID-B254FD2A-669A-4A16-9816-AAC79E983571
DCL List Box and Pop-Up List-Handling	GUID-67E20E58-77B0-434C-B312-055203408B56
DCL Image Tile-Handling	GUID-F9D9DC98-27AB-4BB4-BACC-C5B732110C11
DCL Application-Specific Data-Handling	GUID-3C0BB4C6-D548-45FE-BF16-2D0E2039EC40
DCL Tile Attributes Reference	GUID-5B0C8B10-F968-4A56-B4A5-6A26935B341A
DCL Tiles Reference	GUID-F8F5A79B-9A05-4E25-A6FC-9720216BA3E7
ActiveX Collection Manipulation	GUID-B62C742B-FF80-4D67-A827-EC8E41E31DF8
ActiveX Data Conversion	GUID-042A3895-D875-4FD2-A1E1-A3B565B27DE5
ActiveX Method Invocation	GUID-BE140F3B-7FA8-47A5-B90E-A97CBFDF7501
ActiveX Object-Handling	GUID-6FB61071-0F0D-40EF-A873-7C3561F03C29
ActiveX Property-Handling	GUID-656BFB33-0F62-42BF-A211-20C481C946A9
Curve Measurement	GUID-4684C76F-02F7-4989-AA53-C886E528350A
Dictionary (ActiveX)	GUID-F5F376FF-9BC2-476A-9EA9-1455D28DE264
Drawing Objects	GUID-DF1CB039-265D-4D17-A86E-EF966DC4C141
Reactors	GUID-B83B512E-CEF6-43C5-9099-398999E254AF
VLX Namespace	GUID-5784FC6F-82DD-4459-879B-6EC3BD5E88D1
Namespace Communication	GUID-F5DD5472-1695-4CA3-9110-E6008E87BB89
Express Tools — Entity and Selection	GUID-676F8B1B-7C23-4480-9DD4-19D0836EAE08
Express Tools — File	GUID-05EB15E4-F0ED-4AC7-885D-94B4F6DE9F91
Express Tools — Initialization File / Registry	GUID-9F789A1F-D7E1-4655-87C3-906B908AF5B8
Express Tools — String	GUID-4CE7D81F-BFE6-4221-B9FE-EC7D7680546E
Express Tools — System	GUID-6BCE925F-D717-4BCF-88F8-9F6FBBE7427A
Express Tools — User Interface and Help	GUID-F2967662-438E-425A-BFCC-BD7526074C13

Construct individual page URLs as <https://help.autodesk.com/cloudhelp/2026/ENU/AutoCAD>

6.3 6.3 Per-letter index GUIDs

Letter	GUID
Non-alphabetic	GUID-EBDC1072-48BF-4204-80B1-73430DBF58E1
A	GUID-985B33C7-0A4B-4FBE-8DC2-7EE2E673CC49
B	GUID-72F902B6-5665-48C3-9F49-6ADC643FFCB0
C	GUID-24064CFF-9E44-4A3A-905F-CF6B7B56BCDB
D	GUID-76EE3904-E89B-441A-84FD-C0AB3289742C
E	GUID-512FDA63-ABA7-4C3C-A2DF-F700C440A7AA
F	GUID-4A2DC31F-E543-48A8-BC0A-0AF5E29A3034
G	GUID-88320FBD-B49E-445A-A5C1-C2ABA98341BC
H	GUID-B3916684-1429-4213-875B-E52A5FE5102A
I	GUID-D7758C19-D29B-4362-BBC8-1273294B1756
L	GUID-923D3AAC-6A65-4266-B270-8E90C3F423ED
M	GUID-00EF1664-BBAD-476D-A31C-2F6DD90C1BB4
N	GUID-FFE879C9-9C1B-49E7-8182-313FFD7CAC6B
O	GUID-4C431C76-F9F6-482C-B7C6-B63EB0614DDE
P	GUID-088AD4D0-7088-4465-8CA5-57E8453480E7
Q	GUID-D576D1AC-4A40-442F-AFC7-05638821EAAA
R	GUID-F141C324-736E-4A9F-9DD3-B445C93984FA
S	GUID-FAD2CDD5-50CD-4328-B391-C898AAD4E401
T	GUID-424BAAF1-69C6-4DBF-9948-8B09DF0A40A9
U	GUID-BF6E5B5D-781C-4DF1-85ED-3109430C76CE
V	GUID-26FF2BE2-B5BF-4DD1-A617-36727E51F516
W	GUID-4A800AF5-42E5-486F-9AD5-47CE0CAFCED0
X	GUID-AC778C63-ADF7-4D92-88FE-034FB4C131DD
Z	GUID-1628C9F6-677D-4C2A-A7C7-51C543B994E7

(There are no public J, K, or Y letter pages because no AutoLISP function name starts with those letters.)

7 7 Spec-only Entries and Issues

The following 14 spec dictionary entries do not appear on the Autodesk per-letter pages. All but one are concept-level entries that are correct as spec material:

- Reader-syntax classes: `Integer Literal`, `Real Literal`, `String Literal`,

`Symbol`, `List`, `Dotted Pair`, `Quote` — these have no individual vendor function page.

- Type entries: `ENAME`, `FILE`, `PICKSET`, `REACTOR OBJECT` — documented in vendor prose, not as a function entry.
- Constants: `NIL`, `T`, `PI` — referenced throughout vendor prose.
- `PAUSE` — documented as a special token in user-input prose, not a function.

One issue:

- `AUTLOAD` in the spec is a typo of Autodesk’s `autoload`. Phase 2 must rename the entry.

8 8 Delta — Missing from the Spec

171 symbols documented by Autodesk 2026 are absent from the spec draft. This section partitions them so Phase 2 can attack them in priority order.

8.1 8.1 P0 — Core language essentials (must add first)

These are language fundamentals that any HyperSpec-style reference must contain. Their absence is a structural gap.

- Higher-order: `apply`, `eval`.
- Equality: `eq`, `equal`.
- Arithmetic / numeric predicates: `atan`, `cos`, `sin`, `exp`, `expt`, `log`, `sqrt`, `max`, `min`, `gcd`, `rem`, `numberp`, `1+`, `1-`, `~` (bitwise NOT), `boole`, `logand`, `logior`, `lsh`.
- List access: `cadr`, `caddr`.
- Symbol introspection: `atoms-family`.
- String matching: `wcmatch`.
- List length variant: `vl-list-length`.
- Geometry: `trans`.
- Naming: `autoload` (also fixes the `autoload` typo).

8.2 8.2 P1 — Selection-set / drawing surface

- `ssgetfirst`, `sssetfirst`, `ssmemb`, `ssnamex`, `entmakex`, `entupd`, `regapp`, `xdroom`, `xdsize`.

8.3 8.3 P1 — Display, redraw, and screen control

- `redraw`, `graphscr`, `textpage`, `textscr`, `grclear`, `grdraw`, `grtext`, `grvecs`, `tablet`, `vports`, `setview`.

8.4 8.4 P1 — Application / configuration / environment

- `arx`, `getenv`, `setenv`, `getcfg`, `setcfg`, `getcname`, `layoutlist`, `initcommandversion`, `setfunhelp`, `help`, `menucmd`, `menugroup`, `sinvalid`, `acdimenableupdate`.

8.5 8.5 P1 — Memory / debugging

- `alloc`, `expand`, `gc`, `mem`.

8.6 8.6 P1 — Layer state family (14 entries)

- `layerstate-addlayers`, `layerstate-compare`, `layerstate-delete`, `layerstate-export`, `layerstate-getlastrestored`, `layerstate-getlayers`, `layerstate-getnames`, `layerstate-has`, `layerstate-import`, `layerstate-importfromdb`, `layerstate-removelayers`, `layerstate-rename`, `layerstate-restore`, `layerstate-save`.

In BricsCAD V26 these are documented under the parallel `vl-layerstates-*` family (10 entries). The spec must record this divergence with cross-references in both directions.

8.7 8.7 P1 — DCL completion

- `term_dialog`, `add_list`, `start_list`, `end_list`, `start_image`, `end_image`, `fill_image`, `slide_image`, `vector_image`, `dimx_tile`, `dimy_tile`, `get_attr`, `client_data_tile`.

8.8 8.8 P2 — Reactor completion

- `vlr-beep-reaction`, `vlr-trace-reaction`, `vlr-notification`, `vlr-set-notification`, `vlr-toolbar-reactor`, `vlr-pers`, `vlr-pers-list`, `vlr-pers-p`, `vlr-pers-release`, `vlr-owners`, `vlr-owner-add`, `vlr-owner-remove`.

8.9 8.9 P2 — ActiveX completion

- `vlax-tmatrix`, `vlax-typeinfo-available-p`.

8.10 8.10 P2 — Visual LISP registry (Windows / Mac plist)

- `vl-registry-delete`, `vl-registry-descendents`, `vl-registry-read`, `vl-registry-write`.

8.11 8.11 P2 — Visual LISP compile

- `vlisp-compile` — body must record both AutoCAD and BricsCAD signatures, including the BricsCAD `'st / 'lsm / 'lsa` mode symbol (ignored in BricsCAD V26) and the `.des` default output extension.

8.12 8.12 P3 — Express Tools (AutoCAD-Windows only, 57 entries)

These should live in their own chapter — Phase 4 work — so the core chapters do not balloon with Windows-only material.

- `acad_colordlg`, `acad_helpdlg`, `acad_strlsort`, `acad_truecolorcli`, `acad_truecolordlg`, `acad-pop-dbmod`, `acad-push-dbmod`, `acet-ent-geomextents`, `acet-file-attr`, `acet-file-chdir`, `acet-file-copy`, `acet-file-cwd`, `acet-file-dir`, `acet-file-mkdir`, `acet-file-move`, `acet-file-remove`, `acet-file-rmdir`, `acet-help`, `acet-help-trap`, `acet-ini-get`, `acet-ini-set`, `acet-layerp-mark`, `acet-layerp-mode`, `acet-laytrans`, `acet-load-expresstools`, `acet-ms-to-ps`, `acet-ps-to-ms`, `acet-reg-del`, `acet-reg-get`, `acet-reg-machine-prodkey`, `acet-reg-prodkey`, `acet-reg-put`, `acet-reg-user-prodkey`, `acet-ss-drag-move`, `acet-ss-drag-rotate`, `acet-ss-drag-scale`, `acet-str-collate`, `acet-str-equal`, `acet-str-find`, `acet-str-format`, `acet-str-replace`, `acet-str-wcmatch`, `acet-sys-beep`, `acet-sys-command`, `acet-sys-foreground`, `acet-sys-keystate`, `acet-sys-lasterr`, `acet-sys-procid`, `acet-sys-sleep`, `acet-sys-spawn`, `acet-sys-term`, `acet-sys-wait`, `acet-ui-message`, `acet-ui-pickdir`, `acet-ui-progress`, `acet-ui-status`, `acet-ui-txted`.

8.13 8.13 Phase-2 follow-up — BricsCAD V26 entries newly discovered after the live-TOC walk

The Phase-1 revisit on 2026-04-26 (live-DOM walk inside Chrome, extracted to `bricscad-lisp-toc.txt`) revealed 377 BricsCAD V26 LISP TOC leaves that are documented by Bricsys but were absent from the spec. These

are addressed by an expanded chapter 24 (BricsCAD Extensions). The breakdown:

Family	Count	Notes
ExpressTools API Support (<code>acet-*</code>)	208	BricsCAD’s port of AutoCAD ExpressTools
ExpressTools API (<code>acet::*</code> namespace)	8	BricsCAD-only <code>acet::</code> namespace utilities
VLE LISP Function Library (<code>vle-*</code> / <code>vle_*</code>)	105	BricsCAD-only optimiser / utility in
DOSLib for Linux + Mac (<code>dos_*</code> / <code>doc_*</code>)	23	BricsCAD-only port of DOSLib.
Viewport-Layer (<code>vl-vplayer-*</code>)	9	BricsCAD-only viewport-layer properties
Standard / Misc (<code>ads</code> , <code>bpoly</code> , <code>vmon</code> , etc.)	5	AutoLISP-named functions Bricsys o
ActiveX Extension (<code>vlax-*</code>)	4	BricsCAD-only ActiveX additions.
Reactor Extension (<code>vlr-*</code>)	3	BricsCAD-only reactor additions.
VLX Namespace (<code>vlisp-*</code> obsolete shims)	3	Obsolete VLX namespace symbols.
Visual LISP Extension (<code>vl-infp</code> etc.)	3	BricsCAD-only floating-point predic
Error Control (<code>vl-bt</code> , <code>vl-bt-on/off</code>)	3	BricsCAD-only backtrace toggles.
Generic Properties (<code>ispropertyvalid</code> etc.)	2	BricsCAD-only generic-property help
BricsCAD System (<code>bcad\$disable-extended-error</code>)	1	BricsCAD-only error-mode toggle.
Total	377	

The full name → BricsCAD V26 page-URL list lives in `bricscad-stubs-data.tsv` (Phase-2 generator input).

9 9 BricsCAD V26 BricsCAD-Specific Extensions

Bricsys explicitly documents the following 96 LISP symbols as BricsCAD-only extensions. They must each appear in the spec as dictionary entries with availability **BricsCAD**. **since** tags reflect the version field given on the BricsCAD-Specific LISP Functions page when present.

9.1 9.1 Numeric / math (no documented Autodesk equivalent)

- `acos`, `asin`, `atanh`, `ceiling`, `cosh`, `floor`, `log10`, `mod`, `power`, `random`, `round`, `sinh`, `tan`, `tanh`.

Notes:

- `mod` is a BricsCAD modulus function; in Autodesk 2026 `rem` is the analogous primitive.
- `ceiling` / `floor` / `round` have no direct AutoCAD analogue other than `fix` (which truncates).

9.2 9.2 List, string, and control flow extensions

- `position`, `remove`, `string-split`, `until`.

Notes:

- `position` overlaps semantically with Autodesk's `vl-position` but uses a different name.
- `remove` overlaps semantically with `vl-remove`.
- `string-split` has no direct AutoCAD analogue and is a frequent portability stumbling block.
- `until` is a control structure with no Autodesk equivalent.

9.3 9.3 System and process

- `bcad$licenselevels`, `get_diskserialid`, `getpid`, `lisp$enablefastcom`, `lisp$install`, `lisp$version`, `sleep`, `inspector` (since V18.2), `vlisp-optimizer`.

9.4 9.4 DCL extensions and graphical primitives

- `dlg-sysvars` (since V20), `grarc`, `grfill`, `init_dialog`, `redraw_dialog`.

9.5 9.5 vl-* extensions specific to BricsCAD

- Layer-state group (parallel to Autodesk `layerstate-*` family): `vl-layerstates-delete`, `vl-layerstates-getdescription`, `vl-layerstates-getpropertymask`, `vl-layerstates-has`, `vl-layerstates-list`, `vl-layerstates-rename`, `vl-layerstates-restore`, `vl-layerstates-save`, `vl-layerstates-setdescription`, `vl-layerstates-setpropertymask`.
- Annotative-objects group: `vl-annotative-addscale`, `vl-annotative-get`, `vl-annotative-getscales`, `vl-annotative-remove`, `vl-annotative-removescale`, `vl-annotative-reset`, `vl-annotative-scalelist`, `vl-annotative-set`, `vl-annotative-setscales`, `vl-annotative-supported`.
- Sub-entity group (since V23.1): `vl-subent-atpoint`, `vl-subent-select`, `vl-subent-ssadd`, `vl-subent-ssdel`, `vl-subent-ssmemb`.
- Local-undo group (since V24.1): `vl-local-undo-clear`, `vl-local-undo-pop`, `vl-local-undo-push`, `vl-local-undo-reset`, `vl-local-undo-steps`.
- Filesystem and shell: `vl-getcurrentdir`, `vl-setcurrentdir`, `vl-getstartupdir`, `vl-rmdir`, `vl-list-loaded-lisp`.

- Cancel and prompt: `vl-enable-user-cancel`, `vl-hidepromptmenu`, `vl-showpromptmenu`.
- Misc: `vl-getgeomextents`, `vl-string-split`, `vl-vector-project-pointtoentity` (since V20).

9.6 9.6 vla-* and ActiveX extensions

- `vla-collection->list`, `vla-postcommand` (since V22), `_vlax-safearray-mode`, `dictobjname`.

9.7 9.7 VLE (Visual LISP Extensions) optimised intrinsics

- `vle-optimiser` — optimizer toggle.
- `vle-nth0 ... vle-nth9` — optimised positional accessors.
- `vle-cdrassoc` — optimised `(cdr (assoc ...))` idiom.
- `vle-remove-last` — optimised "drop the last element".
- `vle-entget`, `vle-tblsearch`, `vle-dictsearch`, `vle-dictobjname` — optimised drawing-database accessors.
- `vle-ename-valid` — optimised entity-name validity check.
- `vle-file-encoding` — encoding declarator for text-file I/O.

10 10 Cross-Vendor Divergence Items Identified in Phase 1

The following points were noticed during inventory and must be reflected in dictionary entries during Phase 3 / 4:

1. `vlisp-compile` has different mode arguments and different output formats:
 - Autodesk: produces `.fas`, `.vlx` artefacts (per Autodesk reference page).
 - BricsCAD V26: produces `.des` encrypted output; the mode symbol `'st` / `'lsm` / `'lsa` is documented as accepted but ignored.

2. `open` accepts BricsCAD-extended `ccs=` encoding clauses ("`r,ccs=UNICODE`", "`r,ccs=UTF-8`", "`r,ccs=UTF-16LE`", "`w,ccs=UTF-8`", "`w,ccs=UTF-16LE`") that are not part of the Autodesk `open` surface. Autodesk's Unicode story is governed by `LISPSYS` instead; BricsCAD's is per-call.
3. `ssget` and `ssadd` are documented as having BricsCAD-specific enhancements (V9-era compatibility page).
4. `menucmd` is documented in BricsCAD as supporting only the P0 and P1-P16 cursor / pulldown forms, not auxiliary, button, icon, diesel, screen, or tablet menus (V9-era compatibility page).
5. `tan` is BricsCAD-only. AutoCAD users compute it as `(/ (sin x) (cos x))`.
6. AutoCAD's `layerstate-*` family and BricsCAD's `v1-layerstates-*` family overlap in purpose but disagree on names; spec entries must cross-link.
7. AutoCAD's `v1-position` / `v1-remove` vs BricsCAD's `position` / `remove` overlap on functionality; cross-reference required.
8. The Autodesk `v1ax-typeinfo-available-p` function is Windows-only; BricsCAD parity status is unknown (404 on the Bricsys compatibility table).
9. The Autodesk `v1-vbaload` / `v1-vbarun` pair is Windows-only and tied to AutoCAD's VBA module; BricsCAD does not document a parallel VBA surface.
10. `showhtmlmodalwindow` is Windows-only on Autodesk; BricsCAD has no documented equivalent.
11. **`or` returns `T` / `nil`, not the first non-`nil` value.** Discovered during Phase-5 probe-suite preparation against BricsCAD V26 (probe-core's `(setq x (or x default))` idiom silently clobbered `x` with `T`). BricsCAD V26 documents this explicitly on the per-symbol page: "T if any provided argument evaluates as non-NIL; ... NIL otherwise". Autodesk's documentation is consistent with BricsCAD here. Ports from Common Lisp must rewrite `or`-as-default-value idioms to use `if` instead. The spec entry for `or` now records the divergence and the portable idiom; `probe-core.lisp` uses the `if`-guarded form.

12. **BricsCAD V26 atof accepts C99 hex-float syntax** such as "0x1p4" → 16.0. Tested 2026-04-26 against BricsCAD V26 macOS via `scripts/run-probes-direct.sh`; result captured at `results/bricscad/macos/20260426T122808Z/results.sexp`. Locale sensitivity is **not** present on the macOS build (decimal-comma rejected). The conservative `clautolisp` lex model (no hex-float) is therefore strictly narrower than BricsCAD V26.
13. **terpri is zero-arity in BricsCAD V26.** The 2-argument form (`terpri stream`) raises `too few / too many arguments at [TERPRI]`. Use (`princ "\n" stream`) for portable file-handle newlines.
14. **The autolisp-script wrapper redefines printer functions.** The wrapper at `/works/sncf-reseau/src/outils-autolisp/autolisp-script/` redefines `~prin1`, `princ`, `print`, and `prompt` to single-argument helpers that mirror command-line output. Probes that need the 2-argument form (`'prin1 expr stream'`) must use the direct `scripts/run-probes-direct.sh` runner instead.

11 11 Workflow into Phase 2

This document is consumed by the next phase as follows:

1. Phase 2 (index closure) walks the lists in §8 and §9 in priority order, adding stub dictionary entries with the canonical vendor URL and an explicit **Availability** block.
2. Phase 3 (body fill) fetches each per-symbol vendor page and replaces the stub with a full HyperSpec-style entry. Phase 3 will touch the spec body and bump the document version in AutoLISP Spec in line with the AGENTS.md rule.
3. Phase 4 (divergence reconciliation) addresses §10 explicitly.
4. Phase 5 (closure) finishes the items in `specification-resolution-plan.org` (`atof`, `atoi` lex edges, `vl-member-if-not`, `exit` vs `quit`, printer surface).

12 12 Status

- Document type: working artefact, not part of the normative spec.
- License: CC-BY-SA, matching the spec.

- Phase: 3 of 5 complete.
- Date: 2026-04-26.

12.1 Phase progress

- Phase 1 (vendor inventory, delta, divergence list): done.
 - 2026-04-26 follow-up: re-walked the BricsCAD V26 LISP TOC inside Chrome via Claude-in-Chrome MCP, recovering 941 in-scope BricsCAD V26 LISP entries (vs. the 96 that were visible to the original raw-HTTP fetch). Inventory tables in §3.2, §4 and §8.13 updated.
- Phase 2 (index closure): done; document version bumped to 0.2.0.
 - 2026-04-26 follow-up: 377 newly-discovered BricsCAD-only stubs added to chapter 24 under a new "Phase-2 Follow-up Catalogue" section, grouped by family (ExpressTools API, VLE Library, DOSLib, vlax-, vlr-, vlisp-, etc.). Document version bumped to 0.4.0.
- Phase 3 (body fill): done for the original 266 Phase-2 stubs (document version 0.3.0) **and** for the 377 Phase-2-follow-up stubs (Phase-3 follow-up, document version 0.7.0). The follow-up bulk-fetched every BricsCAD V26 per-symbol page in 16-way parallel curl, parsed the chm2web tabular layout (Arguments, Return, Example, Remarks) plus the inline signature, and replaced each chapter-24 stub with the structured body. 316/377 entries received a non-trivial signature; 134 Arguments; 134 Return; 98 Examples; 83 Remarks. The remaining entries kept a sensible default body where the vendor page itself was a one-line description.
- Phase 4 (divergence reconciliation): done. 47 entries gained an explicit '*** See Also' block covering: layerstate-* / vl-layerstates-* family, vl-position / position, vl-remove / remove, expt / power, rem / mod, log / log10 (portable note), 'open' ccs= modes, menucmd surface diff, vlisp-compile output formats, vl-vbaload / vl-vbarun / showhtmlmodalwindow Windows-only, plus portable-idiom notes on BricsCAD-only acos / asin / tan / ceiling / floor / round. 'vlax-typeinfo-available-p' Availability flipped to **Both**. Hygiene: 266 mangled '

‘ directives left over from the Phase-2/3 generators were repaired. Document version bumped to 0.5.0.

- Phase 5 (long-running closure: ‘atof’, ‘atoi’, ‘vl-member-if-not’, ‘exit’ vs ‘quit’, printer surface): done at **documented** level (version 0.6.0) and **now product-tested** against BricsCAD V26 / macOS (version 0.8.0). The 2026-04-26 probe run is at ‘results/bricscad/macos/20260426T122808Z/results.sexp’ and is referenced from each affected spec entry’s ”Tested Behaviour” sub-section. Five additional cross-vendor divergence findings (items 11–14 in §10) emerged from running the probes and are also recorded in the spec body. AutoCAD product-test parity is deferred: the local AutoCAD 2026 install on macOS has no licensed seat available; the dedicated runner ‘scripts/run-probes-direct-autocad.sh’ is in place for the day a seat becomes available.